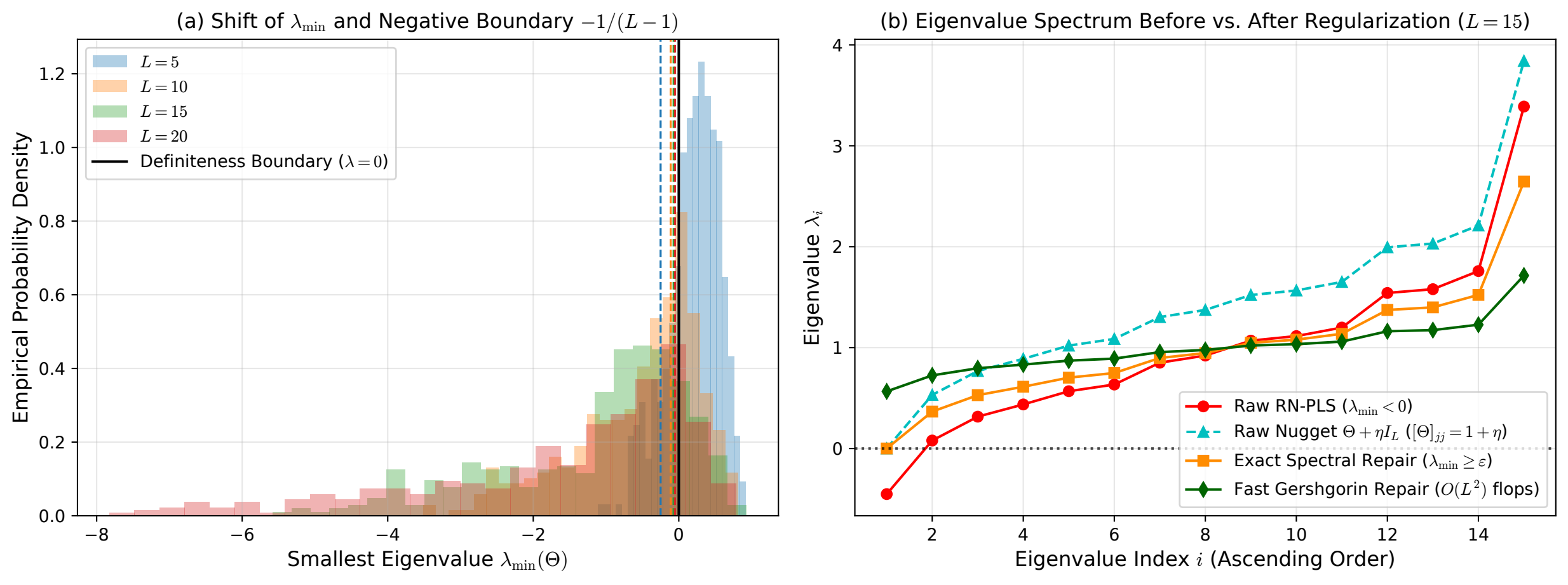
(a) Shift of $\lambda_{\min}$ and Negative Boundary $-1/(L-1)$
$L=5$
$L=10$
$L=15$
$L=20$
Definiteness Boundary ($\lambda=0$)
Empirical Probability Density
Smallest Eigenvalue $\lambda_{\min}(\Theta)$
(b) Eigenvalue Spectrum Before vs. After Regularization ($L=15$)
Raw RN-PLS ($\lambda_{\min}<0$)
Raw Nugget $\Theta+\eta I_L$ ($[\Theta]_{jj}=1+\eta$)
Exact Spectral Repair ($\lambda_{\min}\geq\varepsilon$)
Fast Gershgorin Repair ($O(L^2)$ flops)
Eigenvalue λ_i
Eigenvalue Index i (Ascending Order)